Exam 1 — Foundations under pressure

Questions only

Name: _____ **Date:** _____

Conditions

- 50 questions: 30 single-answer multiple choice and 20 numerical. Each is worth 1 raw practice point.
- Suggested training limit: **150 minutes**. This is not a claim about the actual exam duration or question count.
- Use up to three double-sided A4 sheets, prepared by you, and a local Jupyter notebook. No internet, AI assistance or answer key during the attempt.
- Use Python/NumPy for computation; SciPy is also allowed. You do not need executable OpenCV or PyTorch. Specialized routines are provided when needed.
- For multiple choice, select exactly one of A–D. Attempt every question.
- Unless an exact integer is requested, round the final numerical answer to **two decimal places**. Keep full precision in intermediate steps. Use fractions rather than percentages unless specified.
- Numerical practice grading uses the syllabus rule: within ± 0.01 of the unrounded reference, inclusive. Counts require the exact integer; no partial credit.
- Conventions stated in a question override defaults. Arrays use [row,column] = [y,x]; geometric vectors are columns; image-plane projection is virtual/front-facing; foreground is 1.
- This is original, challenging training material based on the supplied course files, not a lecturer-issued mock or a prediction of the exam.

Solutions are in a separate PDF. Do not open them until you finish.

Part A — Single-answer multiple choice

Q01 — Multiple choice

A camera uses the world-to-camera convention $X_c = RX_w + t$. Which expression gives its optical centre C in world coordinates?

A. $C = -Rt$

B. $C = -R^\top t$

C. $C = R^\top t$

D. $C = t$

Q02 — Multiple choice

An image is resized to half its original width and height, with no crop. Use the ideal continuous-coordinate model, with no pixel-centre offset. Which change is correct?

A. Halve f_x, f_y, skew, c_x and c_y; keep R and t unchanged.

B. Halve only f_x and f_y; leave the principal point unchanged.

C. Halve R and t; leave K unchanged.

D. Halve every entry of K, including $K_{2,2}$, without renormalizing it.

Q03 — Multiple choice

Under an ideal pinhole model, two distinct visible 3D points have exactly the same image coordinates. What must hold?

A. They lie on the same viewing ray from the optical centre.

B. They have the same physical size.

C. They have the same world X and Y coordinates, regardless of camera pose.

D. They have the same depth.

Q04 — Multiple choice

Keep illumination, material and surface orientation fixed. For an ideal Lambertian patch, what is independent of viewing direction?

A. Its outgoing radiance, apart from visibility and camera effects.

B. Its total projected area in the image.

C. Its brightness under any change of light direction.

D. Its surface normal in every coordinate frame.

Q05 — Multiple choice

Twenty calibrated light vectors all lie in one plane through the origin. In the linear photometric-stereo model $i = Lg$, which claim is correct?

A. Only the albedo is ambiguous; the normal is always unique.

B. Twenty images guarantee a unique g.

C. The component of g perpendicular to that plane is unobservable.

D. Normalizing every light vector makes L full rank.

Q06 — Multiple choice

A photometric-stereo estimate has $g = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^\mathsf{T}$. What is the correct interpretation of $n = \frac{g}{\|g\|}$?

A. The normal must be $\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$.

B. The surface must be infinitely far away.

C. The division produces a valid zero unit normal.

D. The normal is undefined; mask or otherwise handle the zero-reflectance/invalid pixel.

Q07 — Multiple choice

For normalized intensity $0 < r < 1$, apply $s = r^\gamma$ with $0 < \gamma < 1$. What happens?

A. Every output intensity becomes 1.

B. The result is a linear contrast stretch.

C. Interior intensities increase; 0 and 1 remain fixed.

D. Interior intensities decrease; 0 and 1 remain fixed.

Q08 — Multiple choice

Why does a two-pass connected-component labelling algorithm store equivalences between provisional labels?

A. I_t estimates the intensity histogram before labelling.

B. Different provisional labels can later be found to belong to one connected region.

C. I_t needs to assign a different label to every foreground pixel.

D. I_t changes 4-connectivity into 8-connectivity automatically.

Q09 — Multiple choice

Foreground objects contain tiny isolated foreground specks and small background holes. Which sequence directly targets the two defects, assuming suitable structuring elements?

A. Closing, then another dilation.

B. Dilation, then another dilation.

C. Opening, then closing.

D. Erosion, then another erosion.

Q10 — Multiple choice

You implement a sliding dot product with an asymmetric 2D kernel K. What change converts this cross-correlation to mathematical convolution?

A. Flip K horizontally and vertically before the sliding dot product.

B. Divide K by its sum.

C. Negate K without changing its positions.

D. Transpose K only.

Q11 — Multiple choice

Two ideal Gaussian smoothing filters with standard deviations 3 and 4 are applied successively. Ignore sampling and truncation. What is the equivalent standard deviation?

A. 25

B. 12

C. 5

D. 7

Q12 — Multiple choice

A nonzero Fourier coefficient is $a + ib$. In NumPy's argument convention, which pair gives its magnitude and phase?

A. $a + b$, $\arctan2(a, b)$

B. $\sqrt{a^2 + b^2}$, $\arctan2(b, a)$

C. $a^2 + b^2$, $\arctan\left(\frac{b}{a}\right)$ for every a and b

D. $\sqrt{a^2 + b^2}$, $\arctan2(a, b)$

Q13 — Multiple choice

Under valid continuous-domain differentiation and convolution, which operation is equivalent to differentiating a Gaussian-smoothed image?

A. Convolve the image with the corresponding derivative of the Gaussian.

B. Threshold the image and then take a histogram.

C. Convolve the image with the square of the Gaussian.

D. Differentiate the Gaussian's standard deviation instead of a spatial coordinate.

Q14 — Multiple choice

After Canny non-maximum suppression, a chain of weak pixels connects to a strong pixel; another weak chain is disconnected from all strong pixels. Hysteresis should do what?

A. Keep every weak pixel because it exceeds the low threshold.

B. Discard every weak pixel because it is below the high threshold.

C. Keep whichever chain has the greatest number of pixels, regardless of connection.

D. Keep the connected weak chain and discard the disconnected chain.

Q15 — Multiple choice

A local structure tensor has two large eigenvalues. What does this indicate?

A. A guaranteed scale-invariant feature at every resolution.

B. Strong intensity variation in two independent directions: a good corner-like feature.

C. A flat patch.

D. A single straight edge with unconstrained tangential motion.

Q16 — Multiple choice

Why is $y = mx + b$ with vertical-residual ordinary least squares unsuitable for an exactly vertical line?

A. I_t requires equal x and y measurement variances; unequal variances are the reason a vertical line is unrepresentable.

B. The model becomes valid once both coordinate columns are centred, because centring makes the slope finite.

C. Orthogonal-distance fitting handles vertical lines by forcing $m = 0$ in the same $y = mx + b$ representation.

D. A vertical line cannot be represented by a finite m; an orthogonal-distance line model avoids this problem.

Q17 — Multiple choice

For fixed desired success probability and model sample size, what happens to the required RANSAC iteration count when the inlier fraction decreases?

A. I_t decreases because more points are rejected.

B. RANSAC becomes mathematically impossible as soon as inliers are below 50%.

C. I_t increases, often sharply.

D. I_t is unchanged because RANSAC is robust.

Q18 — Multiple choice

In Lucas–Kanade, A has rows $\begin{bmatrix} I_x & I_y \end{bmatrix}$. Which matrix also appears in corner analysis?

A. $A^\mathsf{T}A$: the local second-moment/structure tensor.

B. AA^T, always a 2 by 2 matrix.

C. The intrinsic calibration matrix K.

D. The homography's last row.

Q19 — Multiple choice

A Lucas–Kanade window crosses the boundary between two independently moving objects. Which assumption is most directly violated?

A. The image contains any gradients at all.

B. The image has more than one colour channel.

C. The camera has a finite focal length.

D. A single common flow vector for all pixels in the window.

Q20 — Multiple choice

Which property is preserved by every nonsingular affine image transformation, but not by every projective transformation?

A. All Euclidean lengths.

B. All angles.

C. The location of the origin.

D. Parallelism of lines.

Q21 — Multiple choice

Two views of a static scene are taken by rotating a camera about exactly the same optical centre. The scene is not planar. Which model still relates the ideal undistorted images?

A. A unique depth-dependent map recovered by inverting the fundamental matrix.

B. No global model can exist unless the scene is planar.

C. One homography, independent of scene depth.

D. A 2D affine map for every possible rotation angle and calibration, without a projective denominator.

Q22 — Multiple choice

Why impose $\|h\| = 1$ when solving the homogeneous DLT problem $\min_h \|Ah\|^2$ for a homography?

A. Because all nine entries of H must equal 1.

B. Without a scale constraint, $h = 0$ is a trivial minimizer and does not define a homography.

C. Because every image point must have unit Euclidean norm.

D. Because this fixes lens distortion.

Q23 — Multiple choice

A descriptor's nearest and second-nearest match distances are almost equal. Why does Lowe's ratio test tend to reject it?

A. Similar distances establish a reliable nearest match; the ratio test should therefore always keep it.

B. The match is ambiguous even if the nearest absolute distance is small.

C. The ratio test rejects it only because the nearest absolute distance must exceed a fixed universal cutoff.

D. I_t has already been proven to be a geometric outlier; descriptor distances directly establish reprojection error.

Q24 — Multiple choice

Two images have identical counts of visual words but different arrangements of those words. A plain Bag-of-Visual-Words histogram will do what?

A. Recover their exact layout from the counts.

B. Represent them identically, because it discards spatial arrangement.

C. Require a different number of histogram bins for each image.

D. Use the image labels as its histogram bin values.

Q25 — Multiple choice

You want a clean estimate of generalization for a BoVW classifier. Which protocol is best?

A. Choose k and vocabulary size using test accuracy, then report that accuracy.

B. Fit the vocabulary on all images, including the test set, to make evaluation easier.

C. Choose every setting using training accuracy alone.

D. Fit the vocabulary on training descriptors; select settings on validation data; evaluate once on untouched test data.

Q26 — Multiple choice

A classifier predicts the majority class for every example and achieves 99% accuracy on a 99:1 class split. What can its minority-class recall be?

A. 0

B. 1

C. 0.99

D. I_t must equal minority-class precision and be positive.

Q27 — Multiple choice

Ignoring boundaries, a stride-1 convolution with shared weights most directly provides which property?

A. Exact translation invariance: the whole feature map never changes after a shift.

B. A separate learned kernel for each image location.

C. Exact invariance to every rotation and scale change.

D. Translation equivariance: shifting the input shifts the feature map.

Q28 — Multiple choice

What is the main proposal-generation difference between Fast R-CNN and Faster R-CNN?

A. Fast R-CNN predicts pixel masks instead of boxes.

B. Fast R-CNN has an RPN and Faster R-CNN removes it.

C. Faster R-CNN uses a learned Region Proposal Network; Fast R-CNN normally relies on external proposals such as Selective Search.

D. Faster R-CNN runs a separate backbone independently on every raw crop.

Q29 — Multiple choice

Which statement correctly distinguishes the usual skip operations of ResNet and U-Net?

A. ResNet adds compatible activations; U-Net concatenates encoder and decoder features across matching scales.

B. Both use elementwise addition with mismatched channel counts; the addition automatically supplies the missing channels.

C. Both normally concatenate along the channel axis, so every skip doubles the number of channels.

D. ResNet concatenates encoder/decoder features; U-Net adds an identity tensor to each residual branch.

Q30 — Multiple choice

Given a homogeneous image point x in image 1 and fundamental matrix F, with Fx nonzero, what does Fx specify in image 2?

A. The unique matching point, including its depth.

B. An epipolar line on which the corresponding point must lie.

C. A viewing-ray depth that becomes metric once F is normalized to unit Frobenius norm.

D. The matching image point after dehomogenizing Fx as if it were a homography output.

Part B — Numerical values

Q31 — Numerical

Use column vectors and the virtual image plane in front of the camera. The camera centre is $C = \begin{bmatrix}1 & 2 & 0\end{bmatrix}^\mathsf{T}$, $X_w = \begin{bmatrix}3 & 5 & 4\end{bmatrix}^\mathsf{T}$, and $X_c = R(X_w - C)$.

```
import numpy as np
C = np.array([1,2,0])
Xw = np.array([3,5,4])
R = np.array([[0,-1,0], [1,0,0], [0,0,1]])
K = np.array([[800,20,320], [0,600,240], [0,0,1]])
```

Projection is $q = KX_c$, followed by $u = \frac{q_0}{q_2}$. Report u in pixels.

Answer: _____

Q32 — Numerical

For one pixel, use $i = Lg$, where light strength and camera gain are already absorbed into L. Compute the least-squares estimate, then report albedo $\rho = \|g\|$.

```
import numpy as np
L = np.array([[0,0,1], [.8,0,.6], [-.8,0,.6],
              [0,.8,.6], [0,-.8,.6]])
i = np.array([.70,.64,.16,.44,.36])
```

Provided routine: `g = np.linalg.lstsq(L, i, rcond=None)[0]` minimizes the sum of squared residuals. `np.linalg.norm(g)` returns the Euclidean norm.

Answer: _____

Q33 — Numerical

Use the height-field convention $n \propto \begin{bmatrix}-p & -q & 1\end{bmatrix}^\mathsf{T}$, where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$. A plane has constant normal proportional to $\begin{bmatrix}2 & -1 & 2\end{bmatrix}$ and $z(0,0) = 5$. Report $z(3,4)$, using unit spatial spacing.

Answer: _____

Q34 — Numerical

A 2-bit image has histogram counts $\begin{bmatrix}1 & 3 & 2 & 2\end{bmatrix}$ for levels $\begin{bmatrix}0 & 1 & 2 & 3\end{bmatrix}$. Use this specific equalization rule: $T(r) = \lfloor 3\,\mathrm{CDF}(r) \rfloor$, with no subtraction of $\mathrm{CDF}_{\min}$. Report $T(1)$ as an exact integer.

Answer: _____

Q35 — Numerical

Close the following 1D binary row with the centred structuring element $\begin{bmatrix}1 & 1 & 1\end{bmatrix}$. Closing means dilation followed by erosion. At each stage retain exactly 11 entries and treat all out-of-range values as 0. Report the number of foreground entries after closing.

$$\begin{bmatrix}0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0\end{bmatrix}$$

Answer: _____

Q36 — Numerical

For the 3 by 3 patch I and kernel K below, report the single valid output of mathematical convolution, not cross-correlation. The kernel is flipped on both axes. Report an exact integer.

$$I = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \\ 2 & 1 & 5 \end{bmatrix}$$

$$K = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 0 \\ 3 & 0 & -2 \end{bmatrix}$$

Answer: _____

Q37 — Numerical

Use the unnormalized DFT $F_k = \sum_{n=0}^{3} f_n \exp\left(-\frac{2\pi i k n}{4}\right)$. For $f = \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}$, report the magnitude of F_1. You can use $\exp(-i\pi/2) = -i$ and $\exp(-i\pi) = -1$.

Answer: _____

Q38 — Numerical

A patch has structure tensor

$$M = \begin{bmatrix} 9 & 3 \\ 3 & 9 \end{bmatrix}.$$

The Harris response is $R = \det(M) - k\,\mathrm{tr}(M)^2$, with $k = 0.04$. Report R.

Answer: _____

Q39 — Numerical

Use $I_x u + I_y v = -I_t$ and one constant flow in the window. Compute the least-squares flow and report u.

```
import numpy as np
A = np.array([[1,0], [0,1], [1,1], [2,0]], dtype=float)
It = np.array([-2,1,-1,-5], dtype=float)
```

Provided routine: `np.linalg.lstsq(A, -It, rcond=None)[0]` minimizes squared residuals.

Answer: _____

Q40 — Numerical

A homography RANSAC uses samples of 4 correspondences, each independently an inlier with probability $w = 0.6$. Report the minimum integer iteration count for success probability at least $p = 0.99$. Use $N = \left\lceil \frac{\log(1-p)}{\log(1-w^4)} \right\rceil$. Assume all-inlier samples are nondegenerate.

Answer: _____

Q41 — Numerical

A homography maps source to target with

$$H = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & -2 \\ 0.1 & 0 & 1 \end{bmatrix}.$$

Use column-vector homogeneous coordinates. Report the target x coordinate of source point (5,4).

Answer: _____

Q42 — Numerical

Backward warping maps a target pixel to source coordinates $(x, y) = (0.25, 0.75)$. Source intensities are $I_{0,0} = 10$, $I_{0,1} = 30$, $I_{1,0} = 50$, $I_{1,1} = 90$. Arrays use $I_{y,x}$. Report the bilinear interpolated intensity. The weights are $(1-x)(1-y)$, $x(1-y)$, $(1-x)y$ and xy, respectively.

Answer: _____

Q43 — Numerical

Rows of the following confusion matrix are actual classes A,B,C; columns are predicted classes A,B,C. Report the F1 score for class B as a fraction, not a percentage.

$$\begin{bmatrix} 18 & 2 & 0 \\ 3 & 12 & 5 \\ 1 & 4 & 15 \end{bmatrix}$$

Answer: _____

Q44 — Numerical

A retrieval query has 4 relevant database images in total. In its returned ranking, relevant images occur at ranks 1,3,5; the fourth is not returned. Use uninterpolated

$$\text{AP} = \frac{\text{sum of precision at each relevant rank}}{\text{total relevant images}}.$$

Report AP as a fraction.

Answer: _____

Q45 — Numerical

Boxes use continuous coordinates $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$, so width $= x_{\max} - x_{\min}$, without a +1 correction. Report IoU for $A = \begin{bmatrix} 0 & 0 & 6 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 & 8 & 5 \end{bmatrix}$.

Answer: _____

Q46 — Numerical

A standard 2D convolution takes an $H \times W \times C_{\text{in}}$ input of $31 \times 29 \times 3$. Kernel height/width $= 5 \times 3$, stride $= 2 \times 2$, padding $= 1 \times 0$ on EACH side, and $C_{\text{out}} = 8$. No dilation. Report the total number of scalar outputs, excluding the batch dimension. Use

$$\text{out} = \left\lfloor \frac{\text{in} + 2 \cdot \text{pad} - \text{kernel}}{\text{stride}} \right\rfloor + 1 \text{ per axis.}$$

Answer: _____

Q47 — Numerical

For the convolution in Q46, report the number of trainable parameters INCLUDING one bias per output channel. I_t is a dense convolution across all 3 input channels; no groups.

Answer: _____

Q48 — Numerical

Flatten a $7 \times 7 \times 16$ feature map and feed it to a fully connected layer with 10 output units. Include one bias per output unit. Report its trainable parameter count.

Answer: _____

Q49 — Numerical

A network applies: a 3×3 convolution with stride 1; a 2×2 pooling layer with stride 2; a 3×3 convolution with stride 1. Report the side length of one final unit's theoretical receptive field in original input pixels. Ignore boundaries. Provided recurrence: $r_{\text{new}} = r_{\text{old}} + (\text{kernel} - 1) j_{\text{old}}$; $j_{\text{new}} = j_{\text{old}} \cdot \text{stride}$; start $r = j = 1$.

Answer: _____

Q50 — Numerical

For $x_2^{\mathsf{T}} F x_1 = 0$, let

$$F = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix},$$

$x_1 = \begin{bmatrix} 12 & 7 & 1 \end{bmatrix}^{\mathsf{T}}$ and candidate $x_2 = \begin{bmatrix} 9 & 9 & 1 \end{bmatrix}^{\mathsf{T}}$. Report the perpendicular distance in pixels from x2 to its epipolar line $l_2 = F x_1$. Provided formula: distance $= \frac{|ax+by+c|}{\sqrt{a^2+b^2}}$, for line $[a, b, c]$.

Answer: _____